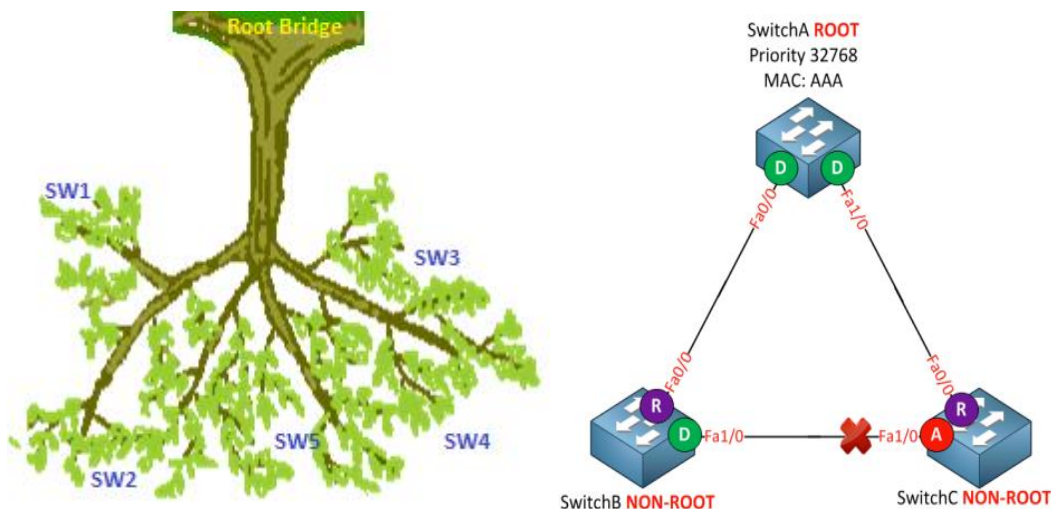


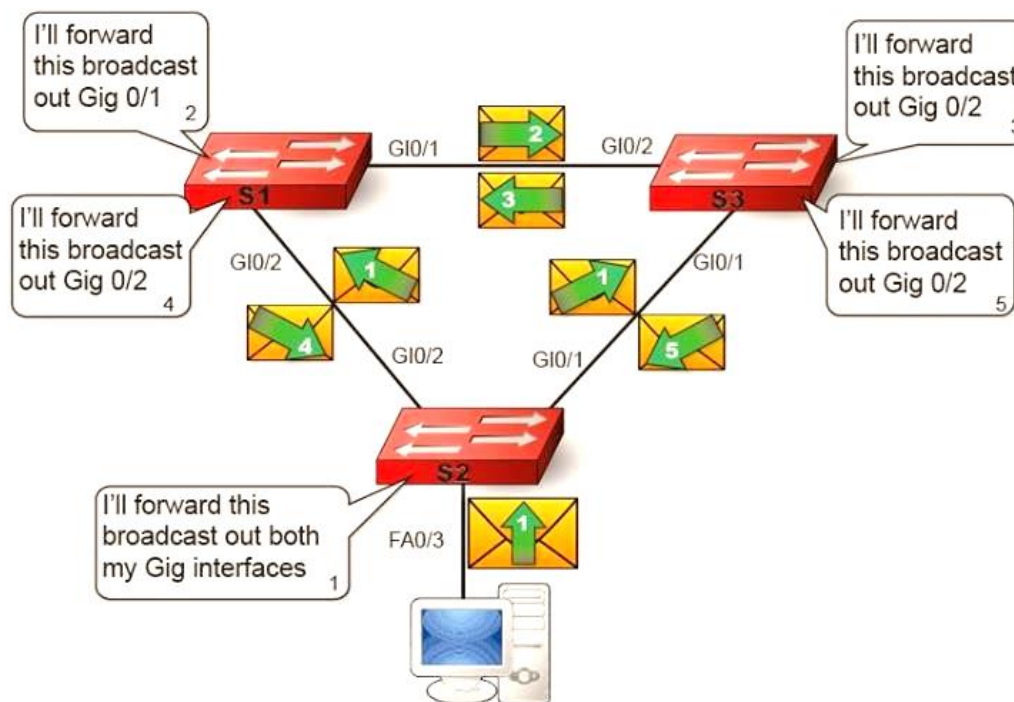
STP (Spanning Tree Protocols):

- o For backup purpose and fault tolerance, we usually create redundant links.
- o Redundant link preventing entire network down situation from single link failure.
- o Redundant link can create network loops that flood down frames in the network.
- o STP automatically removes layer 2 switching loops by shutting down redundant links.
- o Spanning Tree Protocols (STP) is a protocol, it actively monitors all links of the network.
- o Cisco Switch flooding frame unknown unicasting, multicasting and broadcasting.
- o If there are any redundant links available in the network that can cause Layer 2 loops.
- o Layer 2 loops in the network can cause unnecessary resource utilization.
- o Multiple frame transmission, unstable MAC table & unnecessary frame lookup by host.
- o To find a redundant link, it uses an algorithm, known as STA (spanning-tree algorithm).
- o Spanning Tree Protocols use STA (Spanning Tree Algorithm) to prevent Layer 2 loops.
- o STA detecting layer 2 loops and blocks it until the first one link goes down or disconnected.
- o Spanning Tree Protocols is a link management protocol that provides path redundancy.
- o Spanning Tree Protocols (STP) preventing undesirable loops in the whole network.
- o Spanning Tree Protocols is a protocol that runs on switches that helps to solve loops.
- o Spanning Tree Protocols use BPDU in every 2 seconds for preventing Layer 2 loops.
- o Bridge Priority increment uses 4096 because no of VLAN can exist in Cisco switches.
- o Bridge & system ID tie by default so lower MAC address switch selected as Root Bridge.
- o Root Bridge can change timers of Spanning Tree Protocols (STP) and advertise to all.
- o Root Bridge Switch is also responsible for propagating topology change notification.
- o Root Bridge Switch all ports are always designated ports and always in forwarding state.
- o Terms bridge, switch are used interchangeably when discussing Spanning Tree Protocols.



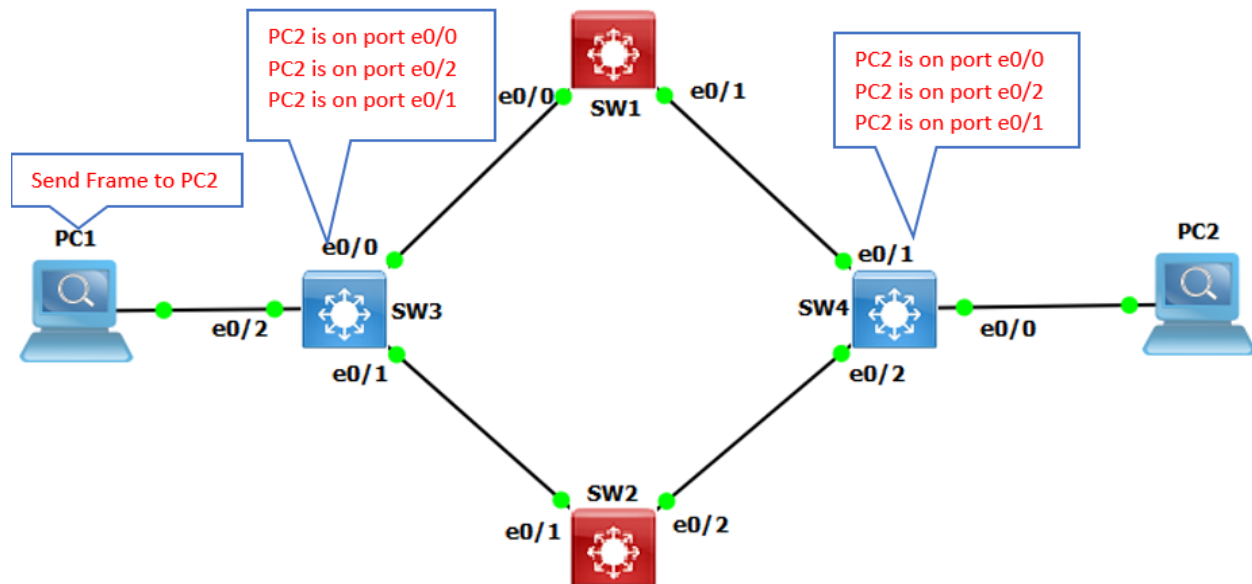
Broadcast Storm:

- o When Switch receives broadcast frames, it continues broadcasting them.
- o The Switches broadcasting them again to its other interfaces or ports.
- o Broadcasting will keep going on forever until you shutdown the network.
- o This phenomenon or situation is called a **broadcast storm of Switches**.
- o The Broadcast storm consumes the entire bandwidth of the network.
- o The Broadcast storm denies bandwidth for normal network traffic.
- o The Broadcast storm can shut down entire network in seconds or less.
- o The Broadcast storms start small but like snowball end up being very big.



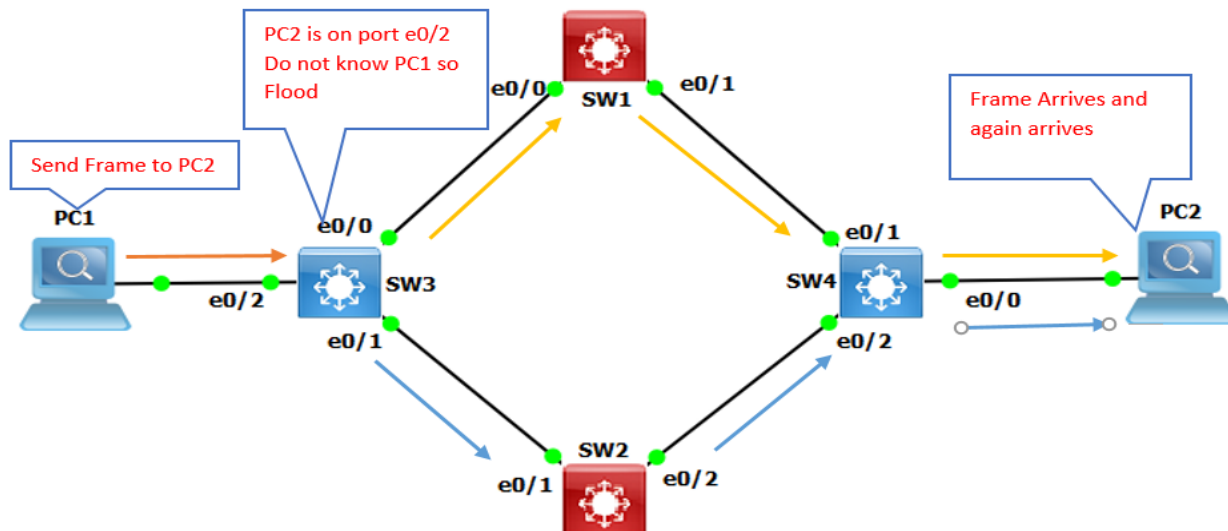
MAC Database Instability:

- o MAC tables are built by examining the source MAC address on a packet received.
- o The source MAC address is tied to the interface or port it was received on.
- o If loop occurs, then same source MAC address could be seen on multiple interfaces.
- o Looped in the network can make MAC Address Table or CAM table unstable.
- o Instability of MAC table causes copies of same frame to be delivered to multiple ports.
- o MAC instability results multiple copies of a frame arrive on different ports of a switch.



Multiple Frame Transmission:

- o Multiple copies of unicast frames may be delivered to destination host.
- o Multiple copies of the same frame can cause unrecoverable errors & issue.



Different Version or Type of Spanning Tree				
Protocol	Standard	Resources Needed	Convergence	Numbers of Trees
STP	802.1D	Low	Slow	One
PVST+	Cisco	High	Slow	One for every VLAN
RSTP	802.1w	Medium	Fast	One
Rapid PVST+	Cisco	Very high	Fast	One for every VLAN
MST	802.1S	Medium or High	Fast	One for multiple VLANs